

Automated Irrigation Controller

```
const int sensorPin = 34;
```

```
const int relayPin = 18;
```

```
bool pumpState = false;
```

```
void setup() {
```

```
    Serial.begin(115200);
```

```
    pinMode(relayPin, OUTPUT);
```

```
    Serial.println("AUTOMATED IRRIGATION CONTROLLER");
```

```
}
```

```
void loop() {
```

```
    int adcValue = analogRead(sensorPin);
```

```
    int moisture = map(adcValue, 4095, 0, 0, 100);
```

```
    if (!pumpState && moisture < 30)
```

```
        pumpState = true;
```

```
if (pumpState && moisture > 45)
  pumpState = false;

digitalWrite(relayPin, pumpState);

Serial.print("Moisture: ");
Serial.print(moisture);
Serial.println("%");

if (pumpState)
  Serial.println("Pump ON");
else
  Serial.println("Pump OFF");

Serial.println("-----");

delay(1000);
}
```

Wokwi Link

<https://wokwi.com/projects/466819048939481089>